🏃 Running Companion — Full Project Plan

Flutter Mobile App · ESP32 Robot Pacer · Firebase Cloud Sync · Generated February 25, 2026

1 · PROJECT OVERVIEW

Running Companion is a mobile + hardware system that lets a runner set a target pace, then have a small 4-wheeled robot physically lead them on an outdoor GPS route — slowing when the runner falls behind, navigating around obstacles, and stopping if the runner gets too far away. Run data is synced to Firebase so history is available across devices.

Mobile App

- Flutter (Android / iOS / web)
- Firebase email/password login
- Pace Workout screen — ar stats
- Run history saved to Firestore
- Connects to robot via WiFi Direct (HTTP)

Robot

- Freenove 4WD ESP32 car kit
- NEO-6M GPS for navigation & speed
- SG90 servo sweeps ultrasonic sensor
- Avoids obstacles, stops if runner lost
- Creates its own WiFi hotspot

2 · CURRENT STATUS

Feature	Status	Notes
Flutter app build & deploy to Pixel 10	Done	Device ID 55301FDCR008RZ, Android 16
App name "Running Companion"	Done	In Flutter + AndroidManifest
Login / Sign-up screen	Done	Email + password UI built
Firestore run history	Done	Save & load run sessions per user
Pace Workout screen	Done	Live stats, pace slider, waypoints
Robot HTTP service (Flutter)	Done	start / stop / status polling
ESP32 firmware — GPS pacing	Done	Follows GPS waypoints at set pace
ESP32 firmware — runner-lost stop	Done	Stops if runner >8 m for 3 s
ESP32 firmware — servo obstacle avoidance	Done	Sweeps left/right, steers around obstacle
Landing website (GitHub Pages)	Done	docs/index.html ready to publish
Firebase configuration	Blocked	Needs `flutterfire configure` — see Step 3
Robot hardware (physical)	Pending	Parts to be ordered
Robot flashed & tested	Pending	After hardware arrives

3 · IMMEDIATE NEXT STEPS

Step A — Configure Firebase (unblocks login + run history)

1. Go to **console.firebase.google.com** → Create project → name it *runner-companion*
2. Authentication → Sign-in method → enable **Email/Password**
3. Firestore Database → Create database → start in **test mode**
4. Add an Android app: package name `com.example.runner_companion_app`
5. In a terminal inside the project folder:

```
dart pub global activate flutterfire_cli
flutterfire configure
```

This replaces `lib/firebase_options.dart` with real credentials. Then rebuild:

```
$env:JAVA_HOME = "C:\Program Files\Eclipse Adoptium\jdk-21.0.10.7-hotspot"
$env:PATH = "$env:JAVA_HOME\bin;$env:PATH"
flutter build apk --release
flutter install -d 55301FDCR008RZ
```

Step B — Order Hardware (~$65 USD)

Item	Search Term	Price
Robot base + ESP32	Freenove 4WD Smart Car Kit ESP32	~$45
GPS module	NEO-6M GPS module Arduino	~$10
Sweep servo	SG90 servo motor micro	~$3
Batteries (×2)	Samsung 25R 18650 or LG HG2 18650	~$12
Jumper wires	Female-to-female dupont jumper wires 20cm	~$5

Step C — Flash the Robot

1. Install **Arduino IDE 2** (arduino.cc)
2. Add ESP32 board: File → Preferences → add `https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json`
3. Install libraries via Sketch → Include Library → Manage Libraries:
   - **TinyGPS++** by Mikal Hart
   - **ArduinoJson** by Benoit Blanchon
   - **ESP32Servo** by Kevin Harrington
4. Open `robot_firmware/runner_pacer/runner_pacer.ino`
5. Select board: **ESP32 Dev Module**, correct COM port
6. Upload ⬆

4 · HARDWARE WIRING GUIDE

GPS Module (NEO-6M → ESP32)

NEO-6M Pin	ESP32 Pin
VCC	3.3V
GND	GND
TX	GPIO 16 (RX2)
RX	GPIO 17 (TX2)

SG90 Servo (front sensor sweep → ESP32)

Servo Wire	ESP32 Pin
Signal (orange)	GPIO 15
VCC (red)	5V
GND (brown)	GND

Ultrasonic Sensors (HC-SR04)

Sensor	TRIG pin	ECHO pin	Purpose
Front	GPIO 12	GPIO 13	Obstacle detection (mounted on servo)
Back	GPIO 4	GPIO 2	Runner distance monitoring

Motor Pins (Freenove kit default)

Motor	FWD	BWD	PWM (speed)
Left	GPIO 26	GPIO 27	GPIO 14
Right	GPIO 33	GPIO 32	GPIO 25

5 · ROBOT BEHAVIOUR LOGIC

Situation	Robot Response
Runner 0–5 m behind	Full target pace
Runner 5–8 m behind	Slow to 60% of target pace
Runner >8 m behind for <3 s	Countdown timer starts
Runner >8 m behind for ≥3 s	**Stops completely** — "Runner Lost" shown in app
Runner comes back within 8 m	Immediately resumes pacing
Obstacle <50 cm ahead	Stops, sweeps servo left & right
One side clear (>80 cm)	Turns and drives around obstacle
Both sides blocked	Reverses slightly, re-scans
Cleared obstacle	Returns to GPS navigation
Final waypoint reached (within 3 m)	Stops, marks workout complete

6 · BATTERY LIFE

The Freenove kit uses **2× 18650 cells**. With ESP32 WiFi hotspot, 4 motors at running pace, servo, and 2 ultrasonic sensors the draw is roughly **1.1–1.5 A**.

Battery quality	Capacity	Runtime
Cheap no-name 18650	~1500 mAh	~30–40 min
Samsung 25R / LG HG2	2500–3000 mAh	**50–75 min**

Speed is measured via **GPS Doppler** — no wheel encoders needed. Accurate to ±0.1 km/h at running speeds.

7 · KEY FILES

File	What it does
`lib/main.dart`	App entry, Firebase init, auth routing, main control page
`lib/firebase_options.dart`	⚠ Placeholder — replaced after `flutterfire configure`
`lib/screens/login_screen.dart`	Email/password login + sign-up UI
`lib/screens/pace_workout_screen.dart`	Live pace workout — stats, waypoints, runner-lost badge
`lib/services/auth_service.dart`	Firebase Auth wrapper (login, signup, logout)
`lib/services/firestore_service.dart`	Save & load run sessions from Firestore
`lib/services/robot_service.dart`	HTTP client to ESP32 robot + status model
`robot_firmware/runner_pacer/runner_pacer.ino`	Full ESP32 firmware with GPS, pacing, obstacle avoidance, runner-lost
`docs/index.html`	GitHub Pages landing website

8 · USEFUL COMMANDS

Build & install APK on Pixel

```
$env:JAVA_HOME = "C:\Program Files\Eclipse Adoptium\jdk-21.0.10.7-hotspot"
$env:PATH = "$env:JAVA_HOME\bin;$env:PATH"
flutter build apk --release
flutter install -d 55301FDCR008RZ
```

Run directly on Pixel (debug)

```
$env:JAVA_HOME = "C:\Program Files\Eclipse Adoptium\jdk-21.0.10.7-hotspot"
$env:PATH = "$env:JAVA_HOME\bin;$env:PATH"
flutter run -d 55301FDCR008RZ
```

Run on Windows desktop

```
flutter run -d windows
```

Run in Chrome

```
flutter run -d chrome
```

Robot WiFi connection

Setting	Value
SSID	RunnerCompanion
Password	runner123
Robot IP	192.168.4.1
Status endpoint	GET http://192.168.4.1/status
Start endpoint	POST http://192.168.4.1/start
Stop endpoint	POST http://192.168.4.1/stop